16 A 100 g block of ice at −10.0 °C is placed into 150 g of water initially at 20.0 °C inside a perfectly insulated calorimeter. What is the final temperature of the mixture and the mass of ice remaining?

Specific heat capacity of water = 4180 J kg^{-1} K^{-1}

Specific latent heat of fusion of ice = 334000 J kg^{-1}

Specific heat capacity of ice = 2100 J kg^{-1} K^{-1}

A 0 K with 31.3 g of ice remaining

B 0 K with 68.7 g of ice remaining

C 0 °C with 31.3 g of ice remaining

D 0 °C with 68.7 g of ice remaining